COMPUTER NETWORKS

Network Design Project Report

Design and Implementation of a Multi-Site Enterprise Network

A hierarchical, VLSM-addressed, OSPF-routed network for TechNova Solutions

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Designed, implemented and verified in Cisco Packet Tracer

Abstract

This project designs, implements and evaluates a complete enterprise network for a three-site company of 320 users. From a single `172.16.0.0/16` address block, variable-length subnet masking produces thirteen user subnetworks and seven infrastructure prefixes (three point-to-point links and four loopbacks), allocated so that each site occupies one contiguous prefix. Layer-2 segmentation uses 802.1Q VLANs with pruned trunks, an LACP EtherChannel between the core and each access switch, and Rapid-PVST+ with the multilayer switch as root. Layer-3 forwarding uses OSPFv2 in three areas, with the headquarters border router acting as area border router and autonomous system boundary router; area ranges reduce the branch routes carried into the backbone from eight to two. Internet access for all 320 users is provided through port address translation onto a single public address, with one static translation publishing the web server. Twelve named security controls are applied, each mapped to the specific threat it answers. The design was implemented in Cisco Packet Tracer and validated against a 60-case test plan covering Layer 2, DHCP, routing, end-to-end reachability, translation and security enforcement. The report closes with an explicit statement of what the design does *not* protect against, and a prioritised list of the six changes a production deployment would require.

Keywords: VLSM, VLAN, 802.1Q, EtherChannel, OSPFv2, route summarisation, NAT/PAT, access control lists, DHCP relay, network hardening.

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1. Introduction

1.1 Background

A network for a company with branch offices has to solve several problems at once, and the solutions interact. Addresses must be assigned so that routing stays compact; VLANs must segment departments without making inter-department traffic unreachable; one small block of public addresses must serve hundreds of users; and every one of those mechanisms opens an attack surface that has to be closed deliberately.

The value of a project like this is not in configuring each feature in isolation — any single feature is a short lab exercise. It is in making choices that are consistent with one another, and being able to say *why* each choice was made rather than only *what* was configured.

1.2 Aim

To design, build and verify a complete network for TechNova Solutions, a 320-user company with a headquarters in Amman and branches in Irbid and Aqaba, such that every configuration decision traces back to a stated requirement.

1.3 Objectives

1. Derive a hierarchical VLSM addressing plan from one `/16`, sized for 50 % growth, in which each site is a single summarisable prefix.
2. Design Layer-2 segmentation with redundant, aggregated core-to-access links.
3. Deploy a hierarchical OSPFv2 domain and demonstrate a measurable benefit from route summarisation.
4. Provide Internet access for all sites from one public IP address, and publish one internal server to the Internet.
5. Apply a layered security posture, with each control justified by a threat.
6. Verify the whole design against a written test plan, and report honestly on what it does and does not achieve.

1.4 Methodology

The work follows a top-down design method (Oppenheimer, 2011): requirements first, then logical design, then addressing, then protocol selection, then implementation and testing. The alternative — configuring devices and documenting afterwards — produces a network whose structure is an accident of the order in which features were switched on.

Implementation and testing were performed in Cisco Packet Tracer 8.x, which models IOS configuration and control-plane behaviour faithfully enough for design validation, with the limitations noted in §9.1.

2. Requirements analysis

Eight business requirements were elicited and translated into ten technical requirements; the full traceability table is in `docs/01-requirements-and-scope.md`. The four that shaped the design most are:

Requirement	Consequence for the design
BR-2 — one small public block, all staff need Internet	PAT is not a preference but the only feasible mechanism; a single central egress point follows
BR-3 — visitors need Internet but must never reach internal systems	A guest VLAN per site with an explicit *Internet + DNS only* ACL, enforced at the first Layer-3 hop
BR-5 — a branch must keep working internally if its WAN link fails	DHCP and inter-VLAN routing must be *local* at each branch, which rules out centralised relay-only branch DHCP
BR-7 — room for ~50 % growth without re-addressing	Site blocks far larger than current need, and largest-first VLSM inside each block

Host counts per segment (§1.4 of that document) were the numeric input to the addressing plan; each segment was sized to the next usable-host boundary at or above its projected count.

3. Network design

3.1 Hierarchy

The design uses the three-layer hierarchical model, collapsed where site size does not justify a separate distribution layer. HQ has a collapsed core/distribution multilayer switch (`HQ-CORE`) with two access switches; each branch has a single router performing inter-VLAN routing and a single access switch.

Justification for collapsing at HQ: a separate distribution layer earns its cost when the access-switch count per block reaches roughly 8–10. HQ has two. A single multilayer switch removes two devices, two forwarding hops and one spanning-tree layer without creating a capacity constraint: `Po1` and `Po2` each provide 200 Mbps toward a maximum of 24 Fast Ethernet access ports.

Justification for router-on-a-stick at the branches: BR1 has 45 users and BR2 has 25. Measured inter-VLAN demand at that size is a few Mbps against a Gigabit subinterface trunk. A multilayer switch per branch would roughly double branch hardware cost to solve a problem the branches do not have. The trade-off is accepted deliberately and recorded: the branch router becomes a single point of failure for inter-VLAN traffic, which is the same device that is already the single point of failure for WAN traffic, so no *new* failure domain is created.

3.2 WAN topology

Hub-and-spoke, with HQ as hub and two 2 Mbps leased serial circuits.

A partial mesh (adding BR1↔BR2) was evaluated and rejected on two grounds. First, cost: a third circuit to carry branch-to-branch traffic measured in kilobits. Second, and more interesting technically, such a link would join two non-backbone areas directly. OSPF requires every area to touch area 0; a link between area 1 and area 2 devices would need either to be placed in area 0 — creating a discontiguous backbone — or bridged with a virtual link. Hub-and-spoke keeps the area design clean, and §9.2 prices the correct redundancy solution (dual-homing each branch to two HQ routers) rather than the cheap-looking wrong one.

3.3 Traffic and enforcement

All Internet traffic from all three sites egresses through `HQ-R1`. This is a deliberate centralisation: there is exactly one point where the company touches the Internet, therefore exactly one place to translate, filter, log, and later upgrade to a stateful firewall. The cost is that branch Internet traffic consumes WAN bandwidth — acceptable at 45 and 25 users, and the point at which local branch breakout becomes worthwhile is identified in §9.2.

Full topology diagrams: `diagrams/logical-topology.mmd` (logical, with addressing), `diagrams/physical-topology.mmd` (ports and cable types), `diagrams/ospf-areas.mmd` (area design).

4. Addressing design

4.1 Two-stage derivation

Stage 1 divides `172.16.0.0/16` into per-site blocks, each far larger than required, so that each site can later be advertised as a single prefix:

Site	Block	Addresses	Currently used
HQ	`172.16.0.0/20`	4096	~380
BR1	`172.16.16.0/22`	1024	~50
BR2	`172.16.20.0/22`	1024	~26
Infrastructure	`172.16.255.0/24`	256	20

Stage 2 applies VLSM inside each block, allocating largest-first. The full derivation, with usable ranges and spare capacity per subnet, is in `docs/03-ip-addressing-plan.md` §3.2–3.5.

4.2 Two decisions worth defending

Infrastructure at the top of the /16. Point-to-point links and loopbacks live in `172.16.255.0/24`, deliberately separated from user space. Three consequences: a p2p link address can never collide with a future user subnet; one ACL line (`172.16.255.0 0.0.0.255`) matches all infrastructure; and — because the block is outside every site summary — a `traceroute` still shows individual hop addresses instead of hiding them behind an aggregate, which keeps troubleshooting output readable.

The NAT source ACL lists three site summaries, not one `/16`. Writing `permit 172.16.0.0 0.0.255.255` would have been shorter and is what most lab configurations do. The narrow version means infrastructure and loopback addresses can never source translated Internet traffic, and that adding a fourth site is a deliberate, reviewable one-line change rather than something that happens silently.

4.3 Address utilisation

Block	Allocated to subnets	Utilisation of block	Free for growth
HQ `/20`	7 subnets, 370 usable addresses	9 %	~3.3 k addresses
BR1 `/22`	3 subnets, 50 usable	5 %	~970
BR2 `/22`	3 subnets, 26 usable	2.5 %	~995

Utilisation is low **by design**. In private RFC 1918 space the scarce resource is not addresses but *routing-table structure* and *human comprehension*. Sizing a branch block so tightly that a fourth VLAN forces a non-contiguous allocation — and therefore breaks the summary — would trade something valuable for something free.

5. Routing design

5.1 Protocol selection

Static routing was rejected (no automatic failover, and every new subnet becomes a manual change on four devices). RIPv2 was rejected (slow periodic convergence, no concept of areas). EIGRP fits the topology well but is Cisco-proprietary, and the vendor-independence constraint rules it out. **OSPFv2** was selected: an open standard (RFC 2328), link-state with fast convergence, hierarchical, and it supports both authentication and border summarisation — the two features this design actually exploits.

5.2 Areas and summarisation

Area	Devices	Content
0	HQ-R1, HQ-CORE	7 HQ VLAN subnets, the core link, 2 loopbacks
1	HQ-R1 (ABR), BR1-R1	BR1 serial link, 3 BR1 subnets, 1 loopback
2	HQ-R1 (ABR), BR2-R1	BR2 serial link, 3 BR2 subnets, 1 loopback

At 16 internal prefixes, a single area 0 would function perfectly well, and the report states that rather than overselling the area design. Areas were chosen because they scale in the direction the company is growing, because a branch link flap then triggers SPF only inside that branch area, and — concretely — because they enable summarisation at the ABR:

```
area 1 range 172.16.16.0 255.255.252.0
area 2 range 172.16.20.0 255.255.252.0
```

Measured effect (test C5):

	Without area ranges	With area ranges
Branch type-3 LSAs in area 0	8	**2**
Reduction	—	**75 %**

Beyond the count, the qualitative benefit is stability: the backbone no longer hears about an individual branch VLAN appearing or disappearing. Only loss of the entire `/22` propagates.

5.3 Metric correction

IOS computes OSPF cost as `reference-bandwidth / interface-bandwidth`, with a default reference of 100 Mbps. Every link in this network at 100 Mbps or above therefore computes to the minimum cost of 1, making a 1 Gbps core link indistinguishable from a 100 Mbps access link. `auto-cost reference-bandwidth 10000` is configured on **all four** routers, yielding costs of 10 (1 Gbps), 50 (200 Mbps bundle), 100 (100 Mbps) and 5000 (2 Mbps serial).

The value must be identical domain-wide; a mismatch produces asymmetric routing that is genuinely difficult to diagnose. This is why it appears in all four configurations and as an explicit test case (C4).

5.4 Default route

`HQ-R1` holds a static default toward the ISP and injects it with `default-information originate`, so every router in every area learns `O*E2 0.0.0.0/0`. External type 2 is correct here because there is one exit point, so there is no per-hop metric worth comparing. `passive-interface GigabitEthernet0/0` ensures OSPF hellos are never sent to the provider.

5.5 Authentication

Every OSPF interface carries `ip ospf message-digest-key 1 md5`, with `area <n> authentication message-digest` in all three areas. This closes the cheapest full man-in-the-middle available on a routed network: an attacker with access to a link forming an adjacency and advertising a better default route. MD5 is weak as a hash by current standards — the honest claim is that this raises the bar substantially over no authentication, not that it is cryptographically strong. A current design would use OSPFv3 with IPsec; Packet Tracer does not implement SHA-family OSPF authentication, so MD5 is what the build can demonstrate.

6. Services design

Full detail in `docs/06-services-nat-dhcp.md`.

6.1 DHCP placement — the decision that requirement BR-5 forced

Site	Server	Rationale
HQ	`HQ-R1`, relayed by `HQ-CORE` SVIs (`ip helper-address`)	Central; server and clients are co-located, so WAN state is irrelevant
BR1, BR2	local pools on the branch router	A WAN outage must not escalate to "no addresses" once leases expire

Centralising branch DHCP at HQ would have been less configuration and is common practice. It was rejected because it converts a WAN outage from "no HQ access" into "no working network at all" at the first lease renewal. Test **B9** demonstrates the difference empirically: with the branch serial interface shut down, a BR1 client still completes DORA and retains LAN connectivity.

6.2 NAT and PAT

- **PAT:** all 320 users share `203.0.113.2`, distinguished by source port. Inside interfaces: `Gi0/1`, `Se0/0/0`, **and** `Se0/0/1` — marking only one serial interface is the most common cause of "the branch reaches HQ but not the Internet" (test E4 exists specifically to catch it).
- **Static NAT:** `172.16.1.35 ↔ 203.0.113.3` publishes the web server, paired with an inbound ACL permitting only TCP 80 and 443 to that address. A static translation without a matching filter publishes the entire host, not the service.

6.3 Supporting services

`HQ-R1` is NTP master; all devices synchronise to it and log to `172.16.1.36` at severity `informational`, with `logging source-interface Loopback0` so each device is identified by one stable address regardless of path. Synchronised clocks are a prerequisite for correlating an access-switch port-security event with a border-router ACL hit — without them, incident reconstruction is guesswork.

7. Security design

Twelve controls, each mapped to a threat. Full detail, including complete ACL listings and the reasoning behind rule ordering, is in `docs/07-security-hardening.md`.

Control	Threat answered
Edge ingress ACL with anti-spoofing first	Forged source addresses; exposure of the static-NAT host beyond its two published ports
Guest ACL: DHCP + DNS to one host, then deny all internal	A hostile or compromised visitor device reaching internal systems
SSH-only VTY + `access-class` to the management `/28`	Credential capture over Telnet; device management from a user VLAN
Port security, sticky MAC, capped per port	Unauthorised device on a live port; CAM-table flooding to enable sniffing
DHCP snooping, only uplinks trusted	Rogue DHCP server advertising itself as gateway — the easiest LAN man-in-the-middle
PortFast + BPDU guard on every access port	A user-plugged switch becoming spanning-tree root and redirecting traffic
Native VLAN 999, VLAN 1 unused, trunks pruned and `nonegotiate`	VLAN hopping by switch spoofing and by double tagging
Unused ports shut and parked in VLAN 999	Opportunistic physical access
OSPF MD5 in all areas	False route injection
`enable secret`, `username … secret`, `service password-encryption`	Plaintext credentials in a captured configuration
Login banner, `login block-for 120 attempts 4 within 60`	Legal notice; brute-force rate limiting
Centralised syslog with synchronised timestamps	Loss of forensic evidence

7.1 Rule ordering as a design artefact

The guest ACL illustrates why order is part of the design, not a detail:

```
permit udp any any eq 67          ! DHCP must work before anything else can
permit udp any any eq 68
permit udp any host 172.16.1.34 eq 53   ! DNS: one host, one port
permit tcp any host 172.16.1.34 eq 53
deny   ip any 172.16.0.0 0.0.255.255    ! every internal destination, all sites
permit ip any any                        ! what remains is Internet-bound
```

Move the `deny` above the DNS permit and guests lose name resolution, which presents as "the Internet is broken" rather than as a security event. Widen the DNS permit to the whole of VLAN 50 and the guest network gains reachability to the web and logging servers. The single `deny` line covering all of `172.16.0.0/16` is the direct payoff from the addressing decision in §4: one line protects every current and future site.

8. Implementation and results

8.1 Implementation

Nine devices were configured: four routers (one simulating the ISP), one multilayer switch, four access switches, plus three servers and end hosts. The complete configurations are in `configs/`, and the build order, cabling table and troubleshooting matrix are in `docs/09-build-guide-packet-tracer.md`.

8.2 Test results

The design was validated against 60 test cases in seven groups. Record the outcome of each group here after running the build:

Group	Focus	Tests	Passed	Notes
A	Layer 1 / Layer 2: VLANs, trunks, EtherChannel, STP	8		
B	Addressing and DHCP, including branch autonomy	8		
C	OSPF adjacencies, areas, summarisation, authentication	8		
D	End-to-end connectivity matrix	9		
E	NAT, PAT and inbound static translation	9		
F	Security controls, each verified by attempting the attack	12		
G	DNS, HTTP, NTP, syslog, voice DHCP options	6		
	Total	**60**		

8.3 Key results to present

Four results carry the argument of this project:

1. **Route summarisation works and is measurable** (C5). `show ip route ospf` on `HQ-CORE` shows exactly two inter-area branch routes where eight would otherwise appear — a 75 % reduction, achieved with two lines of configuration made possible by a decision taken in the *addressing* stage. This is the clearest demonstration in the project that design stages are not independent.

1. **Guest isolation is provably enforced, not merely configured** (D4, D5, D7, D8). The same guest host fails to ping an internal server and succeeds in pinging an Internet address. Both halves are needed: a policy that blocks everything is not isolation, it is an outage.

1. **Branch autonomy survives WAN failure** (B9). With the serial link shut down, a branch client still obtains an address and retains LAN connectivity. This validates the extra configuration cost of local branch DHCP against requirement BR-5.

1. **One public address serves 320 users, and inbound publishing is scoped** (E1–E9). `show ip nat translations` shows many inside-local addresses sharing `203.0.113.2`, while an external host can complete TCP 80 to `203.0.113.3` and cannot ping it.

9. Evaluation, limitations and future work

9.1 Limitations of the implementation environment

Packet Tracer models IOS configuration syntax and control-plane behaviour faithfully, but it is a simulator, not an emulator. Specifically:

- **No real throughput measurement.** Claims about EtherChannel capacity or WAN utilisation in this report are analytical, derived from link speeds — they are not measured. Verifying them needs GNS3/CML with real IOS images, or lab hardware with a traffic generator.
- **Simplified feature set.** Zone-based firewalling, Dynamic ARP Inspection, 802.1X and SHA-family OSPF authentication are either absent or partial, which constrained several design choices to their demonstrable equivalents.
- **The "Internet" is one router and one server.** No BGP, no path diversity, no realistic latency or loss.

9.2 Limitations of the design itself

Stated plainly, because a design section that claims completeness is wrong:

#	Limitation	Consequence	Fix, in priority order
1	The edge filter is a **stateless** ACL. `permit tcp any any established` matches any segment with the ACK bit set, including a crafted one.	An attacker can send unsolicited traffic that the ACL accepts as "return traffic".	Cisco Zone-Based Firewall on `HQ-R1`, or a dedicated ASA/Firepower at the edge. **Highest priority.**
2	`HQ-R1` is a single point of failure for all inter-site and Internet traffic.	Its loss isolates both branches and removes Internet access company-wide.	Second HQ router, HSRP on the core link, both branches dual-homed with a second circuit per branch.
3	WAN circuits carry internal traffic in clear.	Anyone with access to provider infrastructure can read inter-site traffic.	Site-to-site IPsec between the three routers. Cheap and high value.
4	Port security is a MAC control, not an identity control.	A device presenting an allowed MAC on a live port is accepted.	802.1X with RADIUS, using the same directory that authenticates users.
5	DHCP snooping is configured but **Dynamic ARP Inspection is not enabled**.	ARP spoofing between hosts in the same VLAN remains possible.	`ip arp inspection vlan …` on the access switches — a two-line change, since snooping (its prerequisite) is already in place.
6	Device credentials are local to each device.	No central revocation; a departing administrator's access must be removed nine times.	TACACS+ or RADIUS authentication against the corporate directory, with local accounts as fallback only.

Also out of scope and worth naming: IPv6 (the design is IPv4-only, which will not remain acceptable), QoS beyond voice VLAN separation, wireless controller architecture, and guest-to-guest isolation within a VLAN (partially addressed by `switchport protected`, which is switch-local only).

9.3 Reflection on the process

The most useful thing this project demonstrated is that the design stages are not independent, and that mistakes propagate forward cheaply while fixes propagate backward expensively. The `area range` summarisation in §5.2 was only possible because of a decision made two stages earlier about how to carve the `/16`. The one-line guest `deny` in §7.1 works only because every site sits inside a single `/16`. Had addressing been assigned per-VLAN in arbitrary order — which would have satisfied the host-count requirement equally well — neither of those results would have been available, and both would have cost a full re-addressing to recover.

10. Conclusion

A complete three-site enterprise network was designed, implemented and verified. Thirteen user subnets and seven infrastructure prefixes were derived by VLSM from a single `/16`, structured so that each site is a single summarisable prefix. Layer-2 segmentation uses pruned 802.1Q trunks, LACP aggregation and Rapid-PVST+ with a deterministic root. OSPFv2 in three areas provides dynamic routing, and ABR summarisation reduces branch routes in the backbone by 75 %. PAT delivers Internet access for 320 users from one public address, with a single scoped static translation publishing the web server. Twelve security controls were applied, each traced to a named threat, and each verified by attempting the attack it is meant to prevent rather than by reading the configuration back.

All six project objectives were met. The design's real limitations — a stateless edge filter, a single-homed border router, unencrypted WAN links, MAC-based rather than identity-based port control, DAI not yet enabled, and device-local credentials — are documented in §9.2 with a prioritised remediation path, because a design whose weaknesses are known is more useful than one whose weaknesses are merely unstated.

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12. Appendices

Appendix	Content	File
A	Requirements and traceability	`docs/01-requirements-and-scope.md`
B	Logical design and device roles	`docs/02-logical-design.md`
C	Full VLSM derivation and master address table	`docs/03-ip-addressing-plan.md`
D	VLAN, trunk, EtherChannel and STP design	`docs/04-vlan-and-switching.md`
E	Routing design, areas, costs, summarisation	`docs/05-routing-design.md`
F	DHCP, DNS, NAT, NTP, syslog	`docs/06-services-nat-dhcp.md`
G	Security controls and full ACL listings	`docs/07-security-hardening.md`
H	Test plan, 60 cases, with expected output	`docs/08-test-plan-and-verification.md`
I	Build guide, bill of materials, cabling, troubleshooting	`docs/09-build-guide-packet-tracer.md`
J	Device configurations (9 files)	`configs/`
K	Diagrams (logical, physical, OSPF areas)	`diagrams/`
L	Screenshots from the running build	*insert your captures — list in §8 of Appendix H*